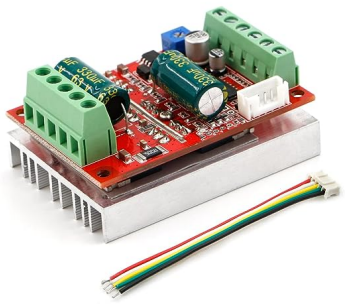
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RioRand 350W 6-60V PWM DC Brushless Electric Motor Speed Controller with Hall

Brand: RioRand

4.1 211 ratings

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- The main board power supply circuit without fuse, can be added(by yourself), the power supply positive and negative terminals will cause permanent damage to some of the chips on the board (not even for a few seconds when the high current)
- Motor output for sampling overcurrent protection, due to the module power supply current are very large. Do not artificially cause a short circuit in the normal operation of the module, a short circuit will damage the tube. Wiring test machine for small-current low-voltage test, successful then through the high-current high-voltage . Bare board module, pay attention to wire insulation when wiring, forbid strong voltage to touch the board.
- Do not connect the obvious and drive module voltage, current, power phase does not match or far from the motor to avoid inexplicable "nonsense" damage.

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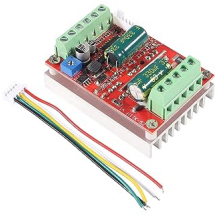
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Product details

Product Dimensions : 2.36 x 1.97 x 1.18 inches; 2.4 Ounces

Item model number : KJL-01

Date First Available : April 23, 2020

Manufacturer : RioRand

ASIN : B087M2378D

Best Sellers Rank: #24,323 in Industrial & Scientific (See Top 100 in Industrial & Scientific)
#12 in Motor Speed Controllers

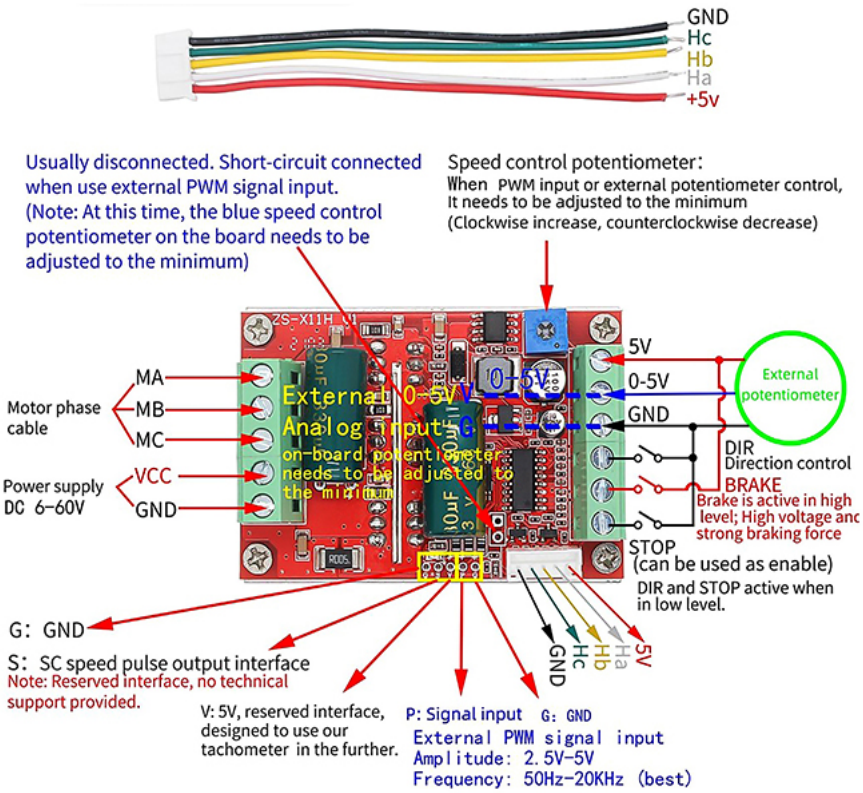
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4.1 211 ratings

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Product Description





350W Brushless Hall DC Motor Driver

Working voltage：6-60V（limit 60V leave a margin for use）

Operating current: rated 16A peak 20A

Power: rated 200-300W peak 350W (leave a margin to strengthen heat dissipation when using large power)

Overcurrent protection: Yes

Note:

1. Because the forward and reverse rotation is no delay hard commutation, and the brake will have a large impact, so you can not operate these two functions at high voltage full speed operation, forced operation may damage the power tube and chip, the need to speed control throttle down to 50% below before operation.

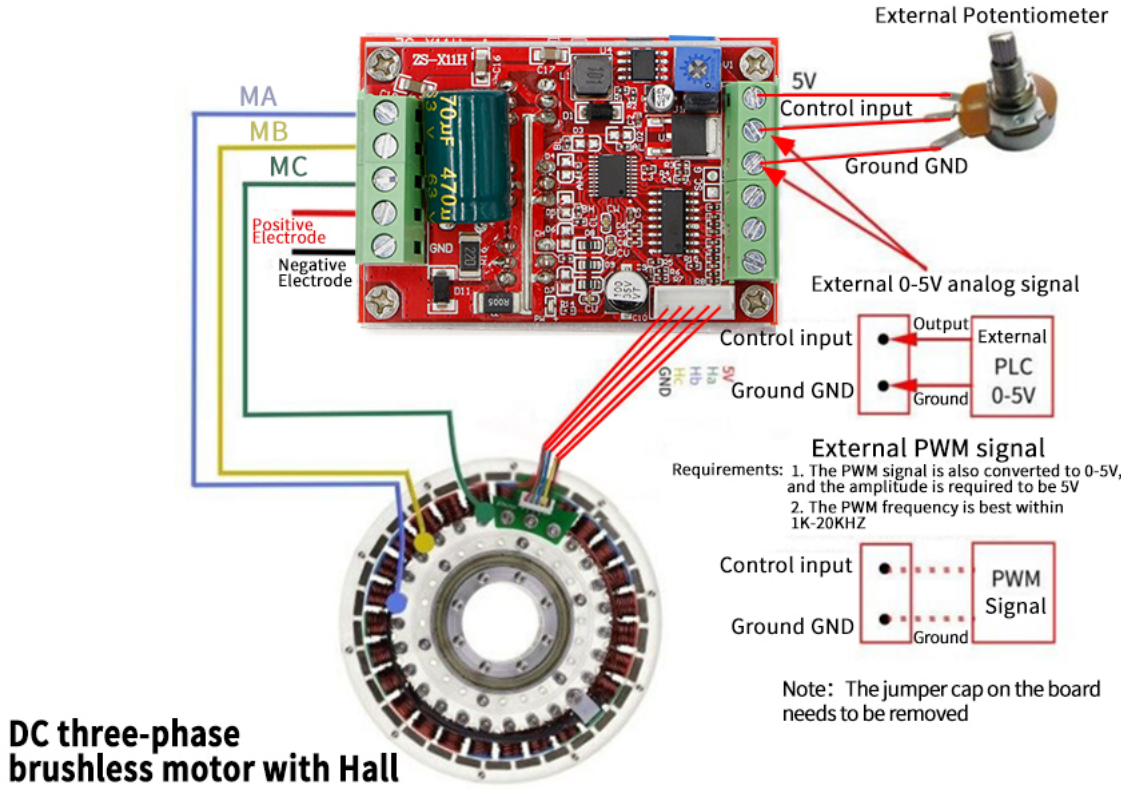
2. The maximum voltage (60V) and maximum power (350W) marked in this module are the limit parameters under specific conditions, so the actual application should leave a margin for use.

If you can't operate this controller, please consider buying it carefully.





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Interface Description:

1.MA MB MC phase line output connected to the motor.

2. 5V GND main board with its own 5V power supply (can be external power supply, the current does not exceed 50MA).

3. VCC GND main power supply (external DC power supply).

4. SC speed pulse signal output (only leave out the interface, no tech+support).

5. DIR direction control, forward and reverse control interface (active low, can be external switch to ground).

6. STOP, stop control interface (active low, can be external switch to ground).

7. BRAKE brake control, brake control interface (active high, external switch to 5V).

8. Speed control speed signal input (integrated potentiometer on the board to regulate speed, can also be external 0-5V analog PWM dual signal input speed regulation).

9. Ha Hb Hc +5V GND Hall signal, power supply input interface, generally with Hall motors have the corresponding 5 lines.

External signal input special instructions:

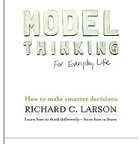
1. From the motherboard P G 2 ports external PWM signal input (the board comes with a potentiometer to the minimum, while shorting the short interface on the board) amplitude 2.5-5V, frequency 50HZ-20KHZ.

2. External analog 0-5V input, from the motherboard 2 terminals sub port input (the board comes with a potentiometer to the minimum)

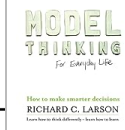
3. External potentiometer speed control, from the motherboard 3 terminal ports correspondingly connected (potentiometer on the board to the minimum)

4. Forward and reverse rotation and stop through a switch to ground (0 potential) low-level valid, but also external MCU to give a low-level signal. Brake through a switch to 5V, high level valid, the front row of terminals, are weak signals directly into the MCU main control, can not touch and contact with high voltage strong current.

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Truth table of motor phase wire and Hall signal wire

		HA	HB	HC	
Corresponding to Hall phase sequence connection	1	ha	hb	hc	The phase wire of the motor can be connected without moving, and only the Hall signal wire can be adjusted.
	2	hc	ha	hb	
	3	hb	hc	ha	
	4	ha	hc	hb	
	5	hb	ha	hc	
	6	hc	hb	ha	

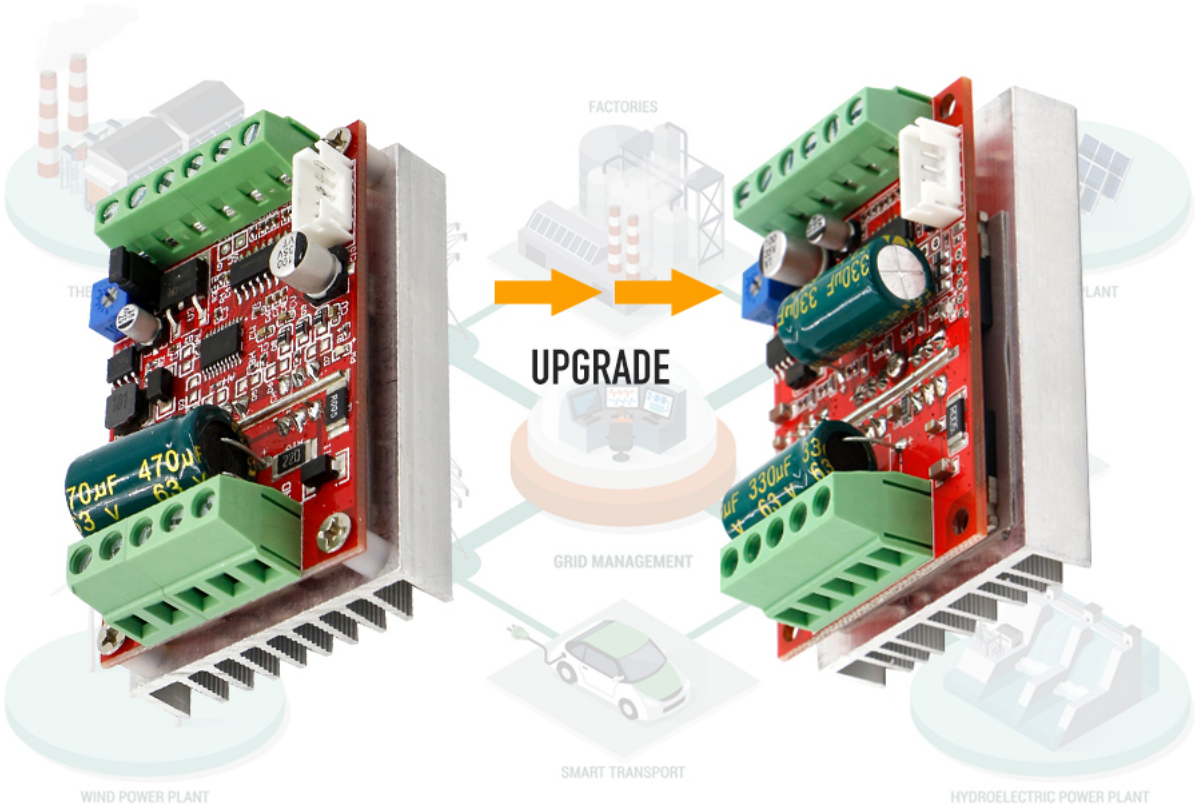
Debugging instructions: Please read and understand before powering on the test machine, this is very important.

- 1. There are generally 5 Hall cables or interfaces on the brushless motor, 2 of which are Hall power lines and 3 are Hall signal lines, which should be distinguished, especially the Hall power lines (the Hall power lines are generally red and black lines). The three Hall signal lines are generally marked with a(Ha) b(Hb) c(Hc), and there are also three port characters such as ha Hb Hc on the driver board, which correspond to each other.
- 2. There are 3 relatively thick phase wires on the motor. There is also a three-phase line interface on the driver board. They are marked with MA(U) MB(V) MC(W) and similar characters, connect them accordingly, and confirm that they are correct. After the main power is turned on, the speed regulation of the potentiometer can be adjusted normally and smoothly. (If the interface is mapped correctly according to the tag, the power supply will not work properly).
- 3. If you do not know the definition of motor phase wire and Hall wire, you can connect the three-phase wire of the motor to the drive phase wire interface at will, and at the same time connect the three signal wires of the Hall to the Hall interface on the drive board. Connect the two hall power wires to each other (one red 5V and one black ground wire of the two hall power wires must be found and connected) and then the first power on, th low voltage is small. Current debugging (adjust the voltage of the constant current power supply to 7-12V current 1-3A if possible), if the motor does not rotate, change the order of the three-phase wires of the motor (connect to any two of the three-phase wires of the motor (wires on the driver board) Swap the wires, and exchange them once every power-on test), until it can rotate smoothly after power-on, which means the wiring is correct. There are 6 different line sequence replacement combinations, please see the picture.
- Please note: When using high voltage, high-speed commutation and braking are not allowed, otherwise the power tube may be damaged by impact.

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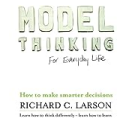




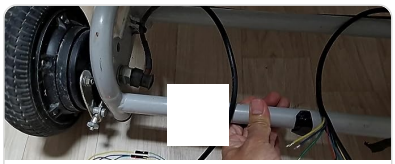
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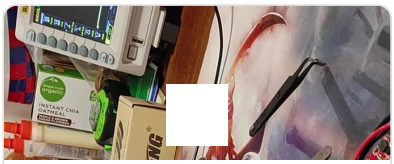


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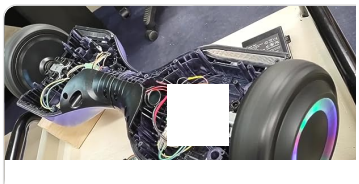
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Customer Review: Good!
MEDITA JIN



0:43

Customer Review: Does exactly what the paper says it does.
TheRealStaciJane



0

Customer Review: Well made, works fine with hoverboard...
S. G.

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Bob Baust

Worked great for me, but WATCH OUT - many questions/answers are for another product

Reviewed in the United States on May 3, 2023

Color: Yellow **Verified Purchase**

Note - many of the questions/answers displayed for this item are actually for the Hall Sensor version so please be careful and don't assume all those answers apply to this module. This listing is for the "Hall-less" version, meaning it's for controlling a 3-phase DC motor that only has 3 wires and does NOT use hall sensor monitoring.

Some errors I noted in the Q/A section:
- some Q/A indicate the module has "soft-start" - it does not, but see my easy hack to add it below*
- some say to "remove the jumper" to use the external analog speed control - there is no jumper on this module, just be sure to set the on-board speed pot to zero (full ccw)
- some say this module will not work with a Hall-less motor - that is WRONG, it is specifically designed to work with hall-less motors.

- there are several different wiring diagrams shown in the listing, but the one that has pictures of the motor and power supply is what I used and seems to be correct.

I used this controller to operate the fan motor from an EGO leaf blower that was trashed because the internal controller died. My fan motor has 3 leads - red, blue, and yellow, which connect to the controller terminals red to red, blue to blue, and yellow to green. I'll use this fan blower with some 18v cordless tool batteries as a sweeper in my shop and patio, and 18v yields plenty of air flow for that purpose.

* I made a simple modification to perform a soft-start function, using a 470uf capacitor and a 1k resistor (see photo and video). Some experimenting proved that the external potentiometer speed control reaches maximum motor speed at about 4.6v, which can be achieved by placing a 1k resistor from the 5v terminal to the analog speed input. By adding the capacitor, the voltage on the speed input slowly rises when power is applied, so that the motor speed ramps up from zero to full speed in about a second, greatly reducing the strain on the power supply, controller, and the motor. If you want a slower start-up you can increase the size of the capacitor, but leave the resistor at 1k. If you increase the resistance above 1000Ω the motor will never reach full speed.

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Helpful Report

Amazon Customer

Works fine with Arduino (Sample code below)

Reviewed in the United States on April 21, 2023

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It took a little bit to figure out the wiring. I was able to crack open my 36V hub motor and clearly read the labels but they didn't correspond to the motor controller.

I did have to short the Jumper.

As far as getting it to work with an Arduino:
- I used an Arduino Nano to test.
- Used the PWM.h library to set the pin frequency to 20khz
- Connected one Arduino pin to "DIR" line and set pin to "HIGH" or "LOW" for Fwd and Rev.
- Used the "G" (Ground), "P" (PWM) and V (5V) connections located by the Jumper to connect to the Arduino. This powered the Arduino as well.

Sample code (I am not the best programmer but it worked):

Read more

2 people found this helpful

Helpful Report

Noah W.

For the price, it's not bad at all. Definitely has some downsides though.

Reviewed in the United States on March 13, 2023

Color: Red **Verified Purchase**

I first bought two of these for controlling hoverboard motors in a project. Wiring them wasn't too difficult, once I figured out the correct phase wires for my motors. For anybody who can't figure this out: take the 6 or 8 screws off the inside of the hoverboard wheel, then use two flat screwdrivers to pry the cover off the wheel. Inside should be a label for each phase wire. I set up my pots and switches for brake/reverse. Don't use the stop function, just use brake. Everything was working as intended. Once proof of concept was achieved, I needed to use an arduino with PWM control to use an RC receiver. This is where my problems began.
First of all, I'm new to arduino so it was a bit of a learning experience. I have other programming language experience so it helped a lot. Once I got my code setup, I jumped my J1 on the board and hooked up everything else. Plugged in power, and the LED instantly turned on then off again. Uh oh. Well, I'll spare the details but two boards later everything is working. I'm still not sure what exactly went wrong the first two times but my best guess is

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Eden Chow

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Reviewed in the United States on June 21, 2023

Color: Yellow **Verified Purchase**

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Helpful Report

Cgessler14

40v Snow Joe snowblower

Reviewed in the United States on June 19, 2023

Color: Yellow **Verified Purchase**

Bought a sunjoe 40v snowblower at a garage sale for $5 with a brushless motor that stopped spinning the auger even tho everything else worked, replaced the electronic motor controller with this and now it's spinning like crazy. Let's go! $20 well spent.

Helpful Report

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Jafet Agustín Jiménez Ramos

funcional

Reviewed in Mexico on February 20, 2022

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es funcional y cumple con su cometido sin embargo al arranque tiene un pequeño retraso de 1 segundo

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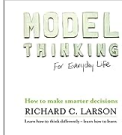
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